

# Local Projection Inference is Simpler and More Robust Than You Think

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January 5, 2021

# Inference on impulse responses

- **Impulse responses** are central objects for causal/counterfactual analysis in macro.
  - Response of  $y_{i,t}$  due to exogenous policy change, at horizons  $h = 1, 2, \dots$
- How to do frequentist **inference**?

① **Vector Autoregression (VAR)**: Iterate on

$$y_t = \sum_{\ell=1}^p A_{\ell} y_{t-\ell} + u_t.$$

Standard errors: delta method or bootstrap.

② **Local Projection (LP)**: Jordà (2005)

$$y_{i,t+h} = \beta_i(h)' y_t + \text{controls} + \xi_{i,t}(h), \quad h = 1, 2, \dots$$

Standard errors: HAC/HAR, since  $\xi_{i,t}(h)$  is serially correlated.

# VAR vs. LP

- Common issues in applied work:
  - ① Persistent data.
  - ② Interest in long impulse response horizons  $h$ .
- **VAR inference**: Standard procedures break down in parts of parameter space.
  - (Near-)unit roots. Phillips (1998); Inoue & Kilian (2020)
  - Delta method singularity. Benkwitz, Neumann & Lütkepohl (2000)
  - Asymptotics: fixed  $h$ . Non-normal limit when  $h = h_T \propto \sqrt{T}$  or  $\propto T$ .  
Wright (2000); Pesavento & Rossi (2006, 2007); Mikusheva (2012)
- **LP inference**: Despite widespread use, no theoretical comparisons with VARs.  
Jordà (2005); Kilian & Kim (2011); Brugnolini (2018)

# Our contributions

- Assume VAR( $p$ ) model.

## ① Lag-augmented LP inference:

$$y_{i,t+h} = \hat{\beta}_i(h)' y_t + \sum_{\ell=1}^p \hat{\gamma}_{i,\ell}(h)' y_{t-\ell} + \hat{\xi}_{i,t}(h), \quad h = 1, 2, \dots$$

## ② Lag-augmented LP inference is uniformly valid over both...

- ❶ DGP. Includes unit root.
- ❷ Horizon  $h$ . Includes  $h = h_T \propto T^\eta$  for  $\eta \in [0, 1)$  (and  $\propto T$  if no unit root).

## ③ Lag augmentation obviates need for HAC/HAR s.e.

- Heteroskedasticity-robust (Eicker-Huber-White) s.e. suffice.
- Simple. No need to choose HAR procedure, tuning parameters.

# Outline

- ① Overview of results: AR(1) case
- ② Comparison with alternative methods: theory and simulations
- ③ General VAR( $p$ ) case
- ④ Conclusion

# Model

- Start with univariate AR(1) model for clarity:

$$y_t = \rho y_{t-1} + u_t, \quad t = 1, 2, \dots, T, \quad y_0 = 0.$$

- Parameter of interest: **impulse response** at horizon  $h$ .

$$\beta(\rho, h) \equiv \rho^h, \quad \rho \in [-1, 1], \quad h \in \mathbb{N}.$$

## Assumption 1: Mean independence

$\{u_t\}$  is strictly stationary, and  $E(u_t \mid \{u_s\}_{s \neq t}) = 0$ .

- Stronger than MDS:  $E(u_t \mid \{u_s\}_{s < t}) = 0$ .
- Satisfied for i.i.d.  $u_t$ . Also allows many types of heteroskedasticity/SV.



## Non-augmented local projection: fragile, HAR s.e.

- AR(1) model implies

$$y_{t+h} = \underbrace{\beta(\rho, h)}_{\equiv \rho^h} y_t + \underbrace{\xi_t(\rho, h)}_{\equiv \sum_{\ell=1}^h \rho^{h-\ell} u_{t+\ell}} .$$

- Non-augmented LP estimator: regress  $y_{t+h}$  on  $y_t$  (no controls).
  - Consistent and asy. normal when  $|\rho| \ll 1$ .
  - Non-normal limit when  $\rho \approx 1$  since  $y_t$  non-stationary.
  - Requires HAR s.e. even when  $|\rho| \ll 1$ , since  $\xi_t(\rho, h)$  serially correlated. HAR inference challenging in small samples. Involves tuning parameters.  
Müller (2007, 2014); Lazarus, Lewis, Stock & Watson (2018)

# Lag-augmented local projection: robust inference

- Lag-augmented LP:

$$\begin{pmatrix} \hat{\beta}(h) \\ \hat{\gamma}(h) \end{pmatrix} \equiv \left( \sum_{t=1}^{T-h} \mathbf{x}_t \mathbf{x}_t' \right)^{-1} \sum_{t=1}^{T-h} \mathbf{x}_t y_{t+h}, \quad \mathbf{x}_t \equiv (y_t, y_{t-1})'.$$

- Would get same  $\hat{\beta}(h)$  if we regressed on  $(u_t, y_{t-1})$ , since  $u_t = y_t - \rho y_{t-1}$ .
- $\hat{\beta}(h)$  has **uniform** normal limit, since

$$y_{t+h} = \beta(\rho, h) \underbrace{u_t}_{\text{stationary}} + \beta(\rho, h+1) \underbrace{y_{t-1}}_{\text{control}} + \xi_t(\rho, h).$$

- $\hat{\gamma}(h)$  non-normal when  $\rho \approx 1$ , but we don't care.



## Lag-augmented local projection: simple standard errors

$$y_{t+h} = \beta(\rho, h)u_t + \beta(\rho, h+1)y_{t-1} + \underbrace{\xi_t(\rho, h)}_{\equiv \sum_{\ell=1}^h \rho^{h-\ell} u_{t+\ell}}$$

- Bonus: lag augmentation simplifies standard errors.
- Leading term in asymptotic expansion:

$$\hat{\beta}(h) \approx \beta(\rho, h) + \frac{\sum_{t=1}^{T-h} \xi_t(\rho, h) u_t}{\sum_{t=1}^{T-h} u_t^2}.$$

- $\xi_t(\rho, h)$  is serially correlated, but **scores**  $\xi_t(\rho, h)u_t$  are not: For  $s < t$ ,

$$E[\xi_t(\rho, h)u_t \xi_s(\rho, h)u_s] = E[\xi_t(\rho, h)u_t \xi_s(\rho, h)] \underbrace{E(u_s \mid u_{s+1}, u_{s+2}, \dots)}_{=0}.$$

- Requires  $E(u_t \mid \{u_s\}_{s>t}) = 0$ . MDS is not enough.

# Lag-augmented local projection: robust inference

- Heteroskedasticity-robust (Eicker-Huber-White) s.e.  $\hat{s}(h)$  suffice. No tuning param's. 
- Define usual confidence interval:

$$\hat{C}(h, \alpha) \equiv \left[ \hat{\beta}(h) - z_{1-\alpha/2} \hat{s}(h), \hat{\beta}(h) + z_{1-\alpha/2} \hat{s}(h) \right].$$

- **Proposition:** This CI is **uniformly** valid.

$$\inf_{\rho \in [-1, 1]} \inf_{1 \leq h \leq \bar{h}_T} P_{\rho} \left( \beta(\rho, h) \in \hat{C}(h, \alpha) \right) \rightarrow 1 - \alpha,$$

for any seq  $\{\bar{h}_T\} \in \mathbb{N}$  such that  $\bar{h}_T/T \rightarrow 0$ .

- Further result: If we restrict  $|\rho| \leq 1 - a$  for  $a > 0$ , then even  $\bar{h}_T \propto T$  is OK.
- Non-normal limit for  $\rho = 1$ ,  $h = h_T \propto T$ .

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# Fragility of AR inference

- Lag-aug LP inference robust to persistence and horizon. Not true for delta method AR. Phillips (1998); Benkwitz et al. (2000); Inoue & Kilian (2002, 2020); Pesavento & Rossi (2007)
- Lag-augmented AR:  $\hat{\beta}_{ARLA}(h) \equiv \hat{\rho}_1^h$ , where  $y_t = \hat{\rho}_1 y_{t-1} + \hat{\rho}_2 y_{t-2} + \hat{u}_t$ .
  - Uniformly  $\sqrt{T}$ -normal limit for fixed  $h$ .
  - Efron bootstrap CI valid at long horizons. Inoue & Kilian (2020)
  - But estimator is inconsistent at horzs  $h = h_T \geq \kappa\sqrt{T}$  when  $\rho \approx 1$ . Confidence interval does not shrink with  $T$  (length can even explode), unlike LP.
- AR grid bootstrap valid at short+long horizons, but *not* intermediate. Computationally intensive. Hansen (1999); Mikusheva (2012)
- In paper: Analytical efficiency rankings of lag-aug/non-aug AR/LP in AR(1) model.

# Simulation study

- Confidence interval procedures:
  - ① Non-augmented AR, delta method s.e. (straw man).
  - ② Lag-augmented AR, Efron bootstrap CI. Inoue & Kilian (2020)
  - ③ Non-augmented LP, percentile-t bootstrap CI, HAR s.e.
  - ④ Lag-augmented LP, percentile-t bootstrap CI, EHW s.e.
- Bootstrap: wild recursive AR design.
- AR(1) model.  $T = 240$ . Nominal confidence level: 90%.
- $u_t \sim N(0, 1)$  i.i.d. (ARCH innovations qualitatively similar.)

| $h$           | Coverage |          |        |          | Median length |          |        |          |
|---------------|----------|----------|--------|----------|---------------|----------|--------|----------|
|               | $AR_d$   | $AR_b^*$ | $LP_b$ | $LP_b^*$ | $AR_d$        | $AR_b^*$ | $LP_b$ | $LP_b^*$ |
| $\rho = 0.50$ |          |          |        |          |               |          |        |          |
| 1             | 0.897    | 0.897    | 0.912  | 0.906    | 0.184         | 0.211    | 0.205  | 0.219    |
| 6             | 0.832    | 0.897    | 0.906  | 0.895    | 0.032         | 0.046    | 0.293  | 0.252    |
| 12            | 0.766    | 0.897    | 0.903  | 0.906    | 0.001         | 0.002    | 0.293  | 0.255    |
| 36            | 0.643    | 0.897    | 0.901  | 0.900    | 0.000         | 0.000    | 0.309  | 0.271    |
| 60            | 0.595    | 0.897    | 0.903  | 0.905    | 0.000         | 0.000    | 0.333  | 0.291    |
| $\rho = 0.95$ |          |          |        |          |               |          |        |          |
| 1             | 0.850    | 0.882    | 0.842  | 0.892    | 0.075         | 0.212    | 0.076  | 0.220    |
| 6             | 0.810    | 0.882    | 0.851  | 0.903    | 0.318         | 1.011    | 0.395  | 0.523    |
| 12            | 0.769    | 0.882    | 0.853  | 0.889    | 0.430         | 1.744    | 0.644  | 0.678    |
| 36            | 0.656    | 0.882    | 0.865  | 0.885    | 0.272         | 6.567    | 0.859  | 0.728    |
| 60            | 0.595    | 0.882    | 0.892  | 0.892    | 0.095         | 23.050   | 0.942  | 0.731    |
| $\rho = 1.00$ |          |          |        |          |               |          |        |          |
| 1             | 0.532    | 0.877    | 0.820  | 0.895    | 0.039         | 0.210    | 0.040  | 0.219    |
| 6             | 0.494    | 0.877    | 0.836  | 0.875    | 0.214         | 1.206    | 0.243  | 0.564    |
| 12            | 0.459    | 0.877    | 0.827  | 0.843    | 0.379         | 2.553    | 0.477  | 0.821    |
| 36            | 0.348    | 0.877    | 0.755  | 0.741    | 0.670         | 21.107   | 1.200  | 1.338    |
| 60            | 0.295    | 0.877    | 0.712  | 0.642    | 0.731         | 161.250  | 1.667  | 1.434    |

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# VAR( $p$ ) model

- VAR( $p$ ) model for  $n$ -dimensional data vector:

$$y_t = \sum_{\ell=1}^p A_{\ell} y_{t-\ell} + u_t, \quad t = 1, \dots, T, \quad y_0 = \dots = y_{1-p} = 0.$$

- Still impose conditional mean independence:  $E(u_t \mid \{u_s\}_{s \neq t}) = 0$ .
- Known lag length  $p$ .
- Reduced-form impulse responses of  $y_{1,t}$  at horizon  $h$ :  $\beta_1(A, h) \in \mathbb{R}^n$ .
- **Parameter of interest**:  $\nu' \beta_1(A, h)$ , where  $\nu \in \mathbb{R}^n$ ,  $\nu \neq 0$ .
  - Simple extension: joint inference on vector  $\beta_1(A, h)$ .
  - Extensions for future work: structural impulse responses, deterministic dynamics.



# Multivariate lag-augmented local projection

- VAR model implies: Jordà (2005); Kilian & Lütkepohl (2017)

$$y_{1,t+h} = \beta_1(A, h)' y_t + \sum_{\ell=1}^{p-1} \delta_{1,\ell}(A, h)' y_{t-j} + \xi_{1,t}(A, h), \quad \xi_{1,t}(A, h) \equiv \sum_{\ell=1}^h \beta_1(A, h - \ell)' u_{t+\ell}.$$

- Lag-augmented LP estimator: regression with  $p$  lags as controls.

$$y_{1,t+h} = \hat{\beta}_1(h)' y_t + \sum_{\ell=1}^p \hat{\delta}_{1,\ell}(h)' y_{t-j} + \hat{\xi}_{1,t}(A, h).$$

- Confidence interval for  $\nu' \beta_1(A, h)$ :

$$\hat{C}_1(h, \nu, \alpha) \equiv \nu' \hat{\beta}_1(h) \pm z_{1-\alpha/2} \underbrace{\hat{s}_1(h, \nu)}_{\text{EHW}}.$$

# Uniformly valid multivariate LP inference

- **Proposition:** Impose As'n "Mean independence". Then the CI  $\hat{C}_1(h, \nu, \alpha)$  is asymptotically uniformly valid over the DGP and horizon  $h$ , provided...
  - 1 The VAR( $p$ ) model can be written in the form  $y_t = \Lambda y_{t-1} + \tilde{y}_t$ , where: **Phillips (1988)**
    - $\tilde{y}_t$  is uniformly stationary VAR( $p - 1$ ) (geometrically decaying IRFs).
    - $\Lambda = \text{diag}(\rho_1, \dots, \rho_n)$  with  $\rho_i \in [-1, 1]$ . **Mikusheva (2012)**
  - 2 Either:
    - i  $h \leq (1 - a)T$  and  $|\rho_i| \leq 1 - a$ ,  $i = 1, \dots, n$ , where  $a > 0$ . OR:
    - ii  $h \leq \bar{h}_T$ , where  $\bar{h}_T/T \rightarrow 0$ .
  - 3 Further regularity conditions: moments, uniform non-singularity of OLS denominator.

# Uniformly valid multivariate LP inference: discussion

- VAR( $p$ ) proof follows AR(1) intuition. Technical challenges:
  - ① Regression scores are not MDS wrt. natural filtration.
  - ② Uniform convergence rates.
  - ③ Uniform bounds on multivariate IRFs.
- We are not aware of other uniformity results that allow multiple (near-)unit roots.
- **Corollary:** Can allow for cointegration among control variables  $y_{2,t}, \dots, y_{n,t}$ .
- Possibly non-normal limit when  $\rho_i \approx 1$  for some  $i$  and  $h \propto T$ .

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# Conclusion

- Local projections already popular in applied work. We provide statistical rationale.
- Lag-augmented LP inference on impulse responses is robust to:
  - ① Persistence of data.
  - ② Length of impulse response horizon.
- Efficiency loss for stationary DGPs at short horizons, but modest in absolute terms.
- Lag augmentation obviates need for HAR s.e. Simple!
- Only known VAR-based methods with comparable robustness are either computationally demanding or can yield very long CIs. Mikusheva (2012); Inoue & Kilian (2020)

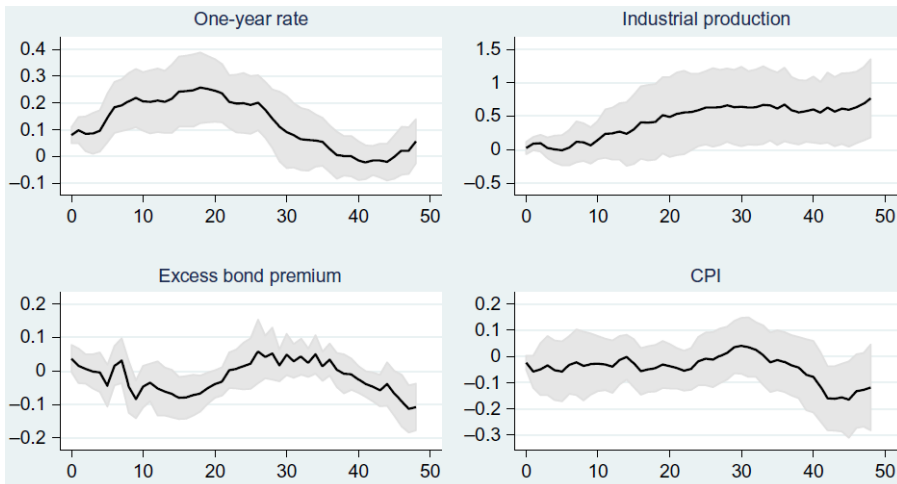
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**Thank you!**

# Appendix

## Example: Ramey (2016) handbook chapter



Gertler-Karadi monetary shock, 90% CI. Sample: 1990m1–2012m6.

Largest horizon  $h = 18\%$  of sample size  $T$ .



## Related literature

- LP vs. VAR: identification, estimation. Plagborg-Møller & Wolf (2020)
- VAR inference.
  - Lag augmentation: Toda & Yamamoto (1995); Dolado & Lütkepohl (1996); Inoue & Kilian (2020)
  - Long horizons: Phillips (1998); Wright (2000); Gospodinov (2004); Pesavento & Rossi (2006, 2007); Mikusheva (2012); I&K (2020)
  - Uniformity: Mikusheva (2007, 2012); I&K (2020)
- LP inference.
  - Pointwise asymptotics: Jordà (2005); Kilian & Lütkepohl (2017); Stock & Watson (2018)
  - Lag augmentation: Dufour, Pelletier & Renault (2006); Breitung & Brüggemann (2020)

**This paper:** uniform LP inference, long+short horizons, simple s.e.

# Heteroskedastic innovations

- Assume  $u_t = \tau_t \epsilon_t$ , where  $\tau_t \geq 0$ .
- Assume  $\epsilon_t$  is i.i.d.,  $E(\epsilon_t) = 0$ .
- Then  $E(u_t \mid \{u_s\}_{s \neq t}) = 0$  if either...
  - ❶  $\{\tau_t\} \perp\!\!\!\perp \{\epsilon_t\}$  (SV).
  - ❷  $\tau_t = f(\epsilon_{t-1}^2, \epsilon_{t-2}^2, \dots)$  and distribution of  $\epsilon_t$  is symmetric (GARCH).



## Lag-augmented local projection: simple standard errors

- Heteroskedasticity-robust (Eicker-White) s.e. suffice:

$$\hat{s}(h) \equiv \frac{(\sum_{t=1}^{T-h} \hat{\xi}_t(h)^2 \hat{u}_t(h)^2)^{1/2}}{\sum_{t=1}^{T-h} \hat{u}_t(h)^2},$$

where

$$\hat{\xi}_t(h) \equiv y_{t+h} - \hat{\beta}(h)y_t - \hat{\gamma}(h)y_{t-1},$$

$$\hat{u}_t(h) \equiv y_t - \hat{\rho}(h)y_{t-1},$$

$$\hat{\rho}(h) \equiv (\sum_{t=1}^{T-h} y_t y_{t-1}) / (\sum_{t=1}^{T-h} y_{t-1}^2).$$

- Readily computed by standard statistical software.
- No tuning parameters.

## AR inference: medium-long horizons

- Suppose  $\rho > 0$  and we use some asy. normal estimator  $\hat{\rho}$ :

$$\sqrt{T}(\hat{\rho} - \rho) \xrightarrow{d} N(0, \tau^2).$$

- Delta method s.e.:

$$\text{se}(\hat{\rho}) \equiv h|\hat{\rho}|^{h-1}\hat{\tau}/\sqrt{T}, \quad \hat{\tau} \xrightarrow{p} \tau.$$

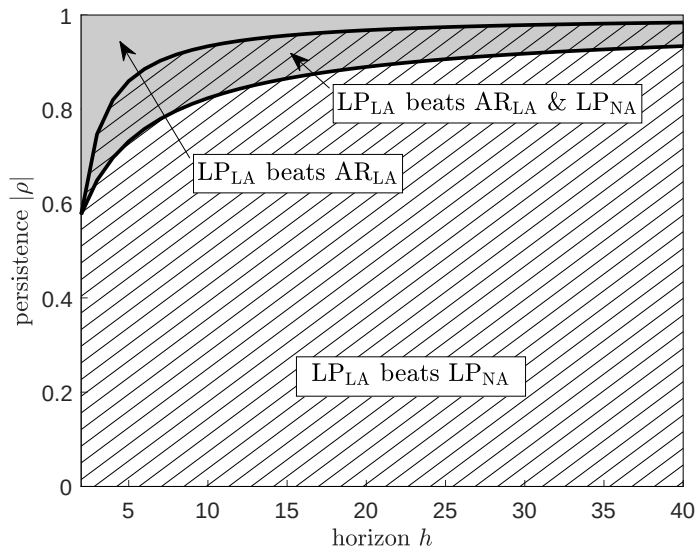
- Then at horizon  $h = h_T = \sqrt{T}$ ,

$$\frac{\hat{\rho}^{h_T}}{\rho^{h_T}} = e^{\sqrt{T}(\log \hat{\rho} - \log \rho)} \xrightarrow{d} e^{N(0, \tau^2/\rho^2)},$$

$$\left| \frac{\sqrt{T}(\hat{\rho}^{h_T} - \rho^{h_T})}{h_T \hat{\rho}^{h_T-1} \hat{\tau}} \right| = \left| \frac{\hat{\rho}}{\hat{\tau}} \left( 1 - \frac{\hat{\rho}^{h_T}}{\rho^{h_T}} \right) \right| \xrightarrow{d} \left| \frac{\rho}{\tau} \left( 1 - N(0, \tau^2/\rho^2) \right) \right|.$$

## Relative efficiency in stationary, fixed-horizon case

- Consider a stationary, homoskedastic VAR model. Fix the horizon  $h$ .
- Then textbook non-augmented AR estimator achieves semiparametric efficiency bound.
- **Ambiguous** ranking of asymptotic variances of inefficient procedures:
  - ① Lag-augmented AR.
  - ② Lag-augmented LP.
  - ③ Non-augmented LP (requires HAC standard errors).
- Next slide: Ranking in homoskedastic AR(1) model, as function of  $(\rho, h)$ .



Trade-off  $LP_{LA}$  vs.  $AR_{LA}$ :

- Non-linear transformation  $\rho^h$ .

Trade-off  $LP_{LA}$  vs.  $LP_{NA}$ :

- Effective regressor  $u_t$  vs.  $y_t$ .
- Serial correl'n of  $\xi_t(\rho, h)y_t$ .

# Simulation results: delta method procedures

| $h$           | Coverage |          |        |          | Median length |          |        |          |
|---------------|----------|----------|--------|----------|---------------|----------|--------|----------|
|               | $AR_d$   | $AR_d^*$ | $LP_d$ | $LP_d^*$ | $AR_d$        | $AR_d^*$ | $LP_d$ | $LP_d^*$ |
| $\rho = 0.50$ |          |          |        |          |               |          |        |          |
| 1             | 0.897    | 0.898    | 0.885  | 0.896    | 0.184         | 0.212    | 0.187  | 0.212    |
| 6             | 0.832    | 0.858    | 0.875  | 0.886    | 0.032         | 0.040    | 0.266  | 0.245    |
| 12            | 0.766    | 0.802    | 0.887  | 0.894    | 0.001         | 0.001    | 0.280  | 0.248    |
| 36            | 0.643    | 0.678    | 0.884  | 0.889    | 0.000         | 0.000    | 0.296  | 0.262    |
| 60            | 0.595    | 0.631    | 0.880  | 0.891    | 0.000         | 0.000    | 0.316  | 0.279    |
| $\rho = 0.95$ |          |          |        |          |               |          |        |          |
| 1             | 0.850    | 0.881    | 0.827  | 0.878    | 0.075         | 0.212    | 0.072  | 0.212    |
| 6             | 0.810    | 0.873    | 0.789  | 0.838    | 0.318         | 0.975    | 0.345  | 0.452    |
| 12            | 0.769    | 0.845    | 0.752  | 0.806    | 0.430         | 1.436    | 0.518  | 0.550    |
| 36            | 0.656    | 0.736    | 0.674  | 0.814    | 0.272         | 1.249    | 0.612  | 0.625    |
| 60            | 0.595    | 0.682    | 0.693  | 0.833    | 0.095         | 0.603    | 0.641  | 0.651    |
| $\rho = 1.00$ |          |          |        |          |               |          |        |          |
| 1             | 0.532    | 0.883    | 0.554  | 0.874    | 0.039         | 0.211    | 0.040  | 0.211    |
| 6             | 0.494    | 0.851    | 0.503  | 0.777    | 0.214         | 1.205    | 0.222  | 0.498    |
| 12            | 0.459    | 0.801    | 0.429  | 0.676    | 0.379         | 2.253    | 0.385  | 0.671    |
| 36            | 0.348    | 0.683    | 0.200  | 0.428    | 0.670         | 5.239    | 0.592  | 0.950    |
| 60            | 0.295    | 0.632    | 0.156  | 0.276    | 0.731         | 6.764    | 0.637  | 0.978    |

# Simulation results: ARCH innovations ( $\alpha_1 = 0.7$ )

| $h$           | Coverage |          |        |          | Median length |          |        |          |
|---------------|----------|----------|--------|----------|---------------|----------|--------|----------|
|               | $AR_d$   | $AR_b^*$ | $LP_b$ | $LP_b^*$ | $AR_d$        | $AR_b^*$ | $LP_b$ | $LP_b^*$ |
| $\rho = 0.50$ |          |          |        |          |               |          |        |          |
| 1             | 0.874    | 0.836    | 0.908  | 0.891    | 0.294         | 0.336    | 0.330  | 0.387    |
| 6             | 0.776    | 0.837    | 0.908  | 0.900    | 0.048         | 0.090    | 0.272  | 0.246    |
| 12            | 0.689    | 0.837    | 0.896  | 0.904    | 0.001         | 0.008    | 0.265  | 0.240    |
| 36            | 0.579    | 0.837    | 0.894  | 0.897    | 0.000         | 0.000    | 0.277  | 0.254    |
| 60            | 0.540    | 0.837    | 0.902  | 0.901    | 0.000         | 0.000    | 0.300  | 0.273    |
| $\rho = 0.95$ |          |          |        |          |               |          |        |          |
| 1             | 0.856    | 0.838    | 0.823  | 0.897    | 0.086         | 0.335    | 0.084  | 0.392    |
| 6             | 0.806    | 0.838    | 0.854  | 0.896    | 0.355         | 1.746    | 0.381  | 0.621    |
| 12            | 0.758    | 0.838    | 0.850  | 0.880    | 0.467         | 3.942    | 0.604  | 0.724    |
| 36            | 0.643    | 0.838    | 0.859  | 0.869    | 0.291         | 64.319   | 0.816  | 0.717    |
| 60            | 0.579    | 0.838    | 0.885  | 0.881    | 0.095         | 1032.604 | 0.900  | 0.711    |
| $\rho = 1.00$ |          |          |        |          |               |          |        |          |
| 1             | 0.560    | 0.839    | 0.841  | 0.896    | 0.041         | 0.330    | 0.040  | 0.386    |
| 6             | 0.513    | 0.839    | 0.859  | 0.879    | 0.223         | 2.035    | 0.240  | 0.686    |
| 12            | 0.468    | 0.839    | 0.845  | 0.854    | 0.391         | 5.510    | 0.473  | 0.902    |
| 36            | 0.352    | 0.839    | 0.752  | 0.731    | 0.669         | 177.260  | 1.170  | 1.384    |
| 60            | 0.294    | 0.839    | 0.697  | 0.640    | 0.729         | 5593.663 | 1.647  | 1.475    |



# VAR parameter space

- Let there be given constants  $a \in [0, 1)$ ,  $C > 0$ , and  $\epsilon \in (0, 1)$ .
- $\mathcal{A}(a, C, \epsilon) \equiv$  space of autoregressive coefficients  $A = (A_1, \dots, A_p)$  such that the associated lag polynomial  $A(L) = I_n - \sum_{\ell=1}^p A_\ell L^\ell$  admits the factorization

$$A(L) = B(L)(I_n - \text{diag}(\rho_1, \dots, \rho_n)L).$$

- $\rho_i \in [a - 1, 1 - a]$  for all  $i = 1, \dots, n$ .
- $B(L)$  is a lag polynomial of order  $p - 1$  with companion matrix  $\mathbf{B}$  satisfying  $\|\mathbf{B}^\ell\| \leq C(1 - \epsilon)^\ell$  for all  $\ell = 1, 2, \dots$



# Uniformly valid LP inference: key proof challenges, AR(1) case

- Study all drifting sequences  $\{\rho_T, h_T\}$ . [Andrews, Cheng & Guggenberger \(2019\)](#)

$$\hat{\beta}(h_T) - \beta(\rho_T, h_T) \propto \sum_{t=1}^{T-h_T} \xi_t(\rho_T, h_T) u_t + \sum_{t=1}^{T-h_T} \xi_t(\rho_T, h_T) \underbrace{[\hat{u}_t(h_T) - u_t]}_{=[\rho_T - \hat{\rho}(h_T)]y_{t-1}}$$

- 1st term:  $E[\xi_t(\rho_T, h_T)^2 u_t^2] \rightarrow \infty$  if  $\rho_T \rightarrow 1, h_T \rightarrow \infty$ .
  - CLT for MDS. Must “reverse time” b/c  $E[\xi_t(\rho, h) u_t \mid \{u_s\}_{s < t+h}] \neq 0$ .
  - Explicitly calculate uniform moment bounds.
- 2nd term: Convergence rate of  $\hat{\rho}(h_T)$  depends on whether  $\rho_T \approx 1$ . [Mikusheva \(2007\)](#)
  - Explicitly calculate uniform moment bounds.

## Proof sketch: reversing time

$$\xi_t(\rho, h) \equiv \sum_{\ell=1}^h \rho^{h-\ell} u_{t+\ell}$$

- Run sum “backwards in time”:

$$\sum_{t=1}^{T-h_T} \xi_t(\rho_T, h_T) u_t = \sum_{t=1}^{T-h_T} \chi_{T,t}, \quad \chi_{T,t} \equiv \xi_{T-h_T-t+1}(\rho_T, h_T) u_{T-h_T-t+1}.$$

- Define filtration

$$\mathcal{F}_{T,t} \equiv \sigma(u_{T-h_T-t+1}, u_{T-h_T-t+2}, \dots).$$

Then  $\chi_{T,t} \in \mathcal{F}_{T,t}$  and  $\mathcal{F}_{T,t} \subset \mathcal{F}_{T,t+1}$  for all  $t$ .

- $\{\chi_{T,t}, \mathcal{F}_{T,t}\}$  is a martingale difference array:

$$E(\chi_{T,t} \mid \mathcal{F}_{T,t-1}) = \xi_{T-h_T-t+1}(\rho_T, h_T) \underbrace{E(u_{T-h_T-t+1} \mid \{u_{T-h_T-t+s}\}_{s>1})}_{=0 \text{ by As'n "Mean independence"}}$$